

# Introduction

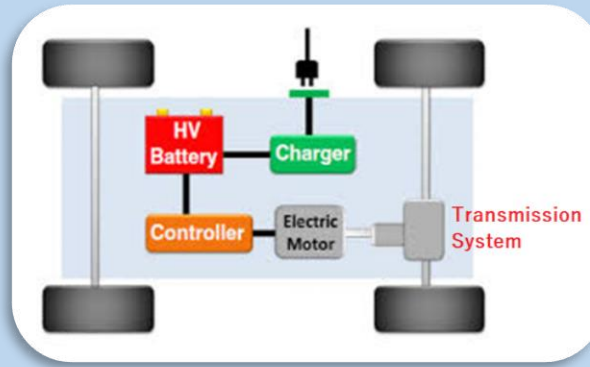
- Urban areas face challenges with high traffic emissions, causing air pollution, noise, and environmental degradation.
- Traditional fossil-fuel vehicles are labor-intensive to maintain, costly to run, and inefficient for daily urban transport.
- Electric cars collect, transfer, and utilize clean battery power, reducing fossil fuel dependence and improving efficiency.
- Electrified transportation is easier to operate and can be powered by renewable energy, supporting sustainable mobility.
- Transitioning to electric vehicles helps cities maintain better environmental standards while lowering overall commuting effort.

# Purposes

- To reduce dependence on fossil-fuel vehicles
- To promote clean transportation
- To study modern technology of Electric Vehicles (EVs)
- To design for small scale transportation
- To fabricate a typical Electric Vehicle (EV)

# Working Principle

- Electric vehicle work on the principle of transforming electric energy into mechanical energy, EVs feature battery and electric motor instead of a conventional fuel engine.



# Specification and Size

| Technical Category                         | Dimension (SI Unit) |
|--------------------------------------------|---------------------|
| Length of the vehicle                      | 2.794 m             |
| Width of the vehicle                       | 1.016 m             |
| Height of the vehicle                      | 1.524 m             |
| Vehicle weight + Four Passengers' weight   | 470 kg              |
| Vehicle max: speed                         | 30 km/hr            |
| Travel distance (Fully charged)            | 20 km               |
| No. of Batteries (12 V, 20.3 Ah) Lead Acid | 5 Nos.              |
| Motor power                                | 3 kW                |
| Wheel diameter                             | 0.381 m             |
| No. of drum brakes                         | 2                   |
| No. of leaf Springs                        | 2                   |



**Technology University  
(Mandalay)**

**Department of Mechanical Engineering**

**“Electric Vehicle (EV Car)”**



## A Quiter Road to Tomorrow

A quiet hum upon the road,  
A lighter step, a gentler load.  
The air is clear, the sky is wide,  
With a quiet future as our guide.



# Fabrication



# General Components



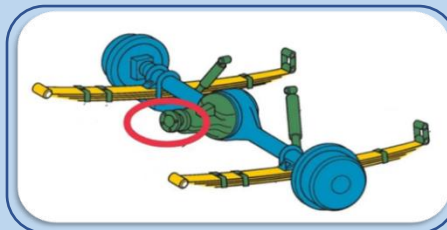
(1) Frame and tyres



(2) Steering System



(3) Braking System



(4) Transmission and Suspension System

